BADR UNIVERSITY IN CAIRO (BUC) — FACULTY OF ENGINEERING

Exercise Sheet (4): Chapter 4 — Hydraulic Pumps | Master Solution & Teaching Guide

Course: MEC 457 / MEC 431 — Fluid Power Systems & Control | **Examiner:** Dr. Ziad Ahmed Ibrahim
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Question 1: Principle of Operation of Positive Displacement Pumps

Working Principle:
A **positive displacement hydraulic pump** operates by mechanically altering the internal volume of sealed fluid chambers:

1. **Suction Phase (Volume Expansion):** As the pumping elements (gears unmeshing, vanes extending, or pistons retracting) move, the volume of the pumping chamber increases. This expansion creates a local partial vacuum (pressure lower than atmospheric), allowing atmospheric or reservoir pressure to push fluid through the suction port into the chamber.
2. **Discharge Phase (Volume Reduction):** The filled chamber is physically isolated from the inlet and swept toward the outlet port. Here, the pumping elements reduce the chamber volume, positively expelling the fluid against external line resistance.

Unlike non-positive dynamic (centrifugal) pumps, a positive displacement pump delivers a nearly constant flow rate that is independent of discharge pressure (governed strictly by speed and displacement), and it will continuously build pressure until the discharge port is blocked!

Question 2: Mathematical Expressions Describing Ideal and Real Pumps

Parameter	Ideal Pump (No Losses)	Real Pump (With Leakage & Friction)
Volumetric Flow Rate (\(Q \))	$$Q_t = V_g \cdot n$$	$$Q = Q_t - Q_L = V_g \cdot n \cdot \eta_v = Q_t - \frac{\Delta P}{R_L}$$
Driving Torque (\(T \))	$$T_t = \frac{V_g \cdot \Delta P}{2\pi}$$	$$T = \frac{T_t}{\eta_m} = \frac{V_g \cdot \Delta P}{2\pi \cdot \eta_m}$$
Input Mechanical Power (\(N_{in} \))	$$N_{in,t} = T_t \cdot \omega = 2\pi n T_t = Q_t \cdot \Delta P$$	$$N_{in} = T \cdot \omega = 2\pi n T = \frac{Q_t \cdot \Delta P}{\eta_m} = \frac{Q \cdot \Delta P}{\eta_t}$$
Hydraulic Output Power (\(N_h \))	$$N_h = Q_t \cdot \Delta P$$	$$N_h = Q \cdot \Delta P = (Q_t \cdot \eta_v) \cdot \Delta P$$
Efficiencies:	• **Volumetric Efficiency:** \(\eta_v = \frac{Q}{Q_t} = 1 - \frac{Q_L}{Q_t} = 1 - \frac{\Delta P}{R_L V_g n} \) • **Mechanical (Hydromechanical) Efficiency:** \(\eta_m = \frac{T_t}{T} = \frac{V_g \Delta P / 2\pi}{T} \) • **Overall (Total) Efficiency:** \(\eta_t = \frac{N_h}{N_{in}} = \frac{Q \cdot \Delta P}{T \cdot \omega} = \eta_v \cdot \eta_m \)	

Question 3: Ideal Displacement Pump Performance Calculation

Given Data:
Geometric displacement: $V_g = 24 \text{ cm}^3/\text{rev} = 24 \times 10^{-6} \text{ m}^3/\text{rev}$
Rotational speed: $n = 1450 \text{ rpm} = \frac{1450}{60} = 24.167 \text{ rev/s}$
Delivery pressure: $P_{out} = 21 \text{ MPa} = 21 \times 10^6 \text{ Pa}$
Inlet pressure: $P_{in} = 150 \text{ kPa} = 0.15 \times 10^6 \text{ Pa}$
Condition: Ideal pump ($\eta_v = 1, \eta_m = 1$).

Step-by-Step Solution:

1. Theoretical Flow Rate (Q_t):
$$Q_t = V_g \cdot n = (24 \times 10^{-6}) \times \left(\frac{1450}{60}\right) = 5.80 \times 10^{-4} \text{ m}^3/\text{s} = \mathbf{34.80 \text{ L/min}}$$

2. Effective Pressure Rise across Pump (ΔP):
$$\Delta P = P_{out} - P_{in} = 21 \times 10^6 - 0.15 \times 10^6 = 20.85 \times 10^6 \text{ Pa} = \mathbf{20.85 \text{ MPa}}$$

3. Increase in Oil Power (ΔN):
$$\Delta N = Q_t \cdot \Delta P = (5.80 \times 10^{-4}) \times (20.85 \times 10^6) = 12,093 \text{ W} = \mathbf{12.09 \text{ kW}}$$

4. Pump Outlet Power (N_{out}):
$$N_{out} = Q_t \cdot P_{out} = (5.80 \times 10^{-4}) \times (21 \times 10^6) = 12,180 \text{ W} = \mathbf{12.18 \text{ kW}}$$

5. Theoretical Driving Torque (T_t):
$$T_t = \frac{V_g \cdot \Delta P}{2\pi} = \frac{(24 \times 10^{-6}) \times (20.85 \times 10^6)}{2\pi} = \frac{500.4}{6.28318} = \mathbf{79.64 \text{ N}\cdot\text{m}}$$ *(Verification: $N_{in} = T_t \cdot 2\pi n = 79.64 \times 2\pi \times 24.167 = 12,093 \text{ W} = \Delta N$).*

Final Answers for Question 3:

Pump Flow Rate: **$Q_t = 34.80$ L/min** | Power Increase: **$\Delta N = 12.09$ kW**

Outlet Power: **$N_{out} = 12.18$ kW** | Driving Torque: **$T_t = 79.64$ N·m**

Question 4: Power Transmission Chain & Causality Relation

Refer to **Quiz Renewable Energy Question 3** (Detailed in Quizzes Master Guide). The flow causality is driven forward by the prime mover speed ($n \to Q \to v$), while pressure causality is driven backward by the external resisting load force ($F \to P \to T \to i$).

Question 5: Cavitation Phenomenon in Displacement Pumps

Physical Mechanism & Damaging Effects:
Cavitation occurs in pumps when local suction pressure drops below the saturated vapor pressure of the oil. Pockets of vapor and released air expand rapidly in the suction chamber. When swept into the high-pressure discharge zone, these

bubbles collapse implosively, generating localized micro-jets with shock pressures up to 1000 MPa and micro-temperatures over 1000°C. This causes severe material pitting/erosion, loud chattering noise, pump vibration, and loss of volumetric flow.

Prevention Guidelines:

- Keep suction pipe flow velocity under $1.0 - 1.2 \text{ m/s}$ (enlarge suction diameter).
- Minimize suction pipe length and avoid unnecessary bends or elbows.
- Never install fine filters on the suction line (use only coarse strainers > 125 µm).
- Mount the hydraulic reservoir above pump level to maintain positive static head.

Question 6: Classification of Hydraulic Pumps

Classification Family	Sub-Types	Displacement Control	Typical Max Pressure
1. Gear Pumps	External Gear, Internal Gear (Gerotor / Crescent)	Fixed Displacement only	Up to 250 bar
2. Vane Pumps	Unbalanced Vane, Balanced Vane (Double chamber)	Fixed or Variable (unbalanced only)	Up to 210 bar
3. Piston Pumps	Axial Piston (Swash plate, Bent axis), Radial Piston	Fixed or Variable (very common)	Up to 350 – 700 bar

Question 7: Flow Pulsation in Axial Piston Pumps

Because each piston executes simple harmonic motion ($v_i = \omega R \sin \theta_i$), the instantaneous flow delivered by the pump is the sum of flows from the pistons connected to the discharge port.

- **Odd Number of Pistons ($z = 7, 9$):** Flow overlaps symmetrically. The delivery pulsation coefficient $\sigma = \frac{Q_{max} - Q_{min}}{Q_{mean}}$ is dramatically lower (approx. $1.5 - 2.5\%$).
- **Even Number of Pistons ($z = 6, 8$):** Two pistons reach top/bottom dead centers simultaneously, creating pulsation coefficients over $8 - 14\%$!

Hence, all high-quality commercial axial piston pumps are built with an ODD number of pistons (almost always 7 or 9 pistons).

Question 8: Internal Gear Pumps (Operation & Characteristics)

An internal gear pump consists of an externally toothed inner spur gear (pinion) meshing with an internally toothed outer ring gear (rotor). A stationary crescent-shaped sealing block fills the clearance between the unmeshed teeth.

- **Operation:** As gears rotate and unmesh, volume expands on the suction side, pulling in oil. The crescent separates suction and pressure zones. Meshing on the discharge side expels the fluid.
- **Characteristics:** Exceptionally quiet operation (< 65 dB), very low flow ripple, compact design, and high volumetric efficiency at medium pressures (up to 250 bar).

Question 9: Pressure-Balanced Vane Pumps

In a standard (unbalanced) vane pump, the rotor is eccentric inside a circular cam ring, concentrating high discharge pressure on one side of the shaft and causing enormous bearing side-loads. In a **pressure-balanced vane pump**, an elliptical cam ring creates **two opposing suction ports and two opposing pressure ports** spaced 180° apart.

Key Advantage: Hydraulic forces on the rotor shaft are completely equal and opposite ($\sum F = 0$). Shaft radial bearing loads are eliminated, allowing operation at higher pressures (up to 210 bar) and extending bearing life tenfold! *Note: Balanced vane pumps can ONLY have fixed displacement.*

Question 10: External Gear Pump Comprehensive Sizing

Given Data:
Speed: $n = 1450 \text{ rpm} = 24.167 \text{ rps}$, Number of teeth: $z = 10$, Module: $m = 4 \text{ mm} = 0.004 \text{ m}$, Tooth width: $b = 22 \text{ mm} = 0.022 \text{ m}$,
Pressure angle: $\alpha = 20^\circ$, Inlet: $P_i = 0.3 \text{ MPa}$, Exit: $P_o = 17 \text{ MPa}$, $\eta_m = 0.89$, $\eta_v = 0.91$.

Step-by-Step Calculations:

1. Geometric Displacement Volume (V_g):
Using Dr. Rabie's standard formula: $$V_g = 2\pi b m^2 (z + \cos^2 \alpha) = 2\pi (0.022) (0.004)^2 (10 + \cos^2 20^\circ)$$ $$\cos 20^\circ = 0.93969 \implies \cos^2 20^\circ = 0.8830$$ $$V_g = 2\pi (0.022) (1.6 \times 10^{-5}) (10.883) = 2.407 \times 10^{-5} \text{ m}^3 = \mathbf{24.07 \text{ cm}^3/\text{rev}}$$

2. Delivery Flow Rates:
$$Q_t = V_g \cdot n = (2.407 \times 10^{-5}) \times 24.167 = 5.817 \times 10^{-4} \text{ m}^3/\text{s} = \mathbf{34.90 \text{ L/min}}$$ $$Q = Q_t \cdot \eta_v = (34.90) \times 0.91 = \mathbf{31.76 \text{ L/min}} = 5.293 \times 10^{-4} \text{ m}^3/\text{s}$$

3. Delivery Pulsation Coefficient (σ):
$$\sigma = \frac{\pi^2 \cos^2 \alpha}{4(z + 1)} = \frac{\pi^2 \times 0.8830}{4(10 + 1)} = \frac{8.7148}{44} = 0.1981 = \mathbf{19.81\%}$$

4. Internal Leakage Flow & Resistance:
$$Q_L = Q_t - Q = 5.817 \times 10^{-4} - 5.293 \times 10^{-4} = 5.24 \times 10^{-5} \text{ m}^3/\text{s} = \mathbf{52.4 \text{ cm}^3/\text{s}}$$ $$\Delta P = P_o - P_i = 17 \times 10^6 - 0.3 \times 10^6 = 16.7 \times 10^6 \text{ Pa}$$ $$R_L = \frac{\Delta P}{Q_L} = \frac{16.7 \times 10^6}{5.24 \times 10^{-5}} = \mathbf{3.187 \times 10^{11} \text{ Pa}\cdot\text{s/m}^3} = \mathbf{318.7 \text{ GNs/m}^5}$$

5. Driving Torque & Input Power:
$$T_t = \frac{V_g \cdot \Delta P}{2\pi} = \frac{(2.407 \times 10^{-5}) \times (16.7 \times 10^6)}{2\pi} = 63.97 \text{ N}\cdot\text{m}$$ $$T_{actual} = \frac{T_t}{\eta_m} = \frac{63.97}{0.89} = \mathbf{71.88 \text{ N}\cdot\text{m}}$$
$$N_{in} = T_{actual} \cdot 2\pi n = 71.88 \times 2\pi \times 24.167 = 10,914 \text{ W} = \mathbf{10.91 \text{ kW}}$$

6. Volumetric Efficiency at Increased Pressure ($P = 210 \text{ bar} = 21 \text{ MPa}$):
New pressure rise: $\Delta P_{210} = 21 \times 10^6 - 0.3 \times 10^6 = 20.7 \times 10^6 \text{ Pa}$.
$$Q_{L,210} = \frac{\Delta P_{210}}{R_L} = \frac{20.7 \times 10^6}{3.187 \times 10^{11}} = 6.495 \times 10^{-5} \text{ m}^3/\text{s}$$ $$\eta_{v,210} = 1 - \frac{Q_{L,210}}{Q_t} = 1 - \frac{6.495 \times 10^{-5}}{5.817 \times 10^{-4}} = 1 - 0.1116 = \mathbf{0.8884 = 88.84\%}$$

Summary for Question 10:

Displacement: **$V_g = 24.07 \text{ cm}^3$** | Pulsation: **$\sigma = 19.81\%$** | Driving Torque: **$T = 71.88 \text{ N·m}$**

Input Power: **$N_{in} = 10.91 \text{ kW}$** | Leakage Resistance: **$R_L = 318.7 \text{ GNs/m}^5$**

Volumetric Efficiency at 210 bar: **$\eta_v = 88.84\%$**

Question 11: Bent Axis Axial Piston Pump

Given Data:
Pistons: $z = 9$, Piston diameter: $d = 10 \text{ mm} = 0.01 \text{ m}$, Pitch circle: $D = 35 \text{ mm} = 0.035 \text{ m}$, Drive speed: $n = 3500 \text{ rpm} = 58.333 \text{ rps}$,
Block inclination angle: $\gamma = 20^\circ$, $P_i = 0.2 \text{ MPa}$, $P = 20 \text{ MPa}$, $\eta_v = 0.92$, $\eta_t = 0.89$, $\eta_h = 1.0$.

Step-by-Step Solution:

(a) Theoretical Flow, Real Flow, Power, and Torque:
Piston stroke: $S = D \sin \gamma = 0.035 \times \sin 20^\circ = 0.01197 \text{ m}$.
Geometric displacement: $$V_g = z \cdot \left(\frac{\pi}{4} d^2\right) \cdot (D \sin \gamma) = 9 \times \frac{\pi}{4} (0.01)^2 \times 0.01197 = 8.462 \times 10^{-6} \text{ m}^3 = \mathbf{8.46 \text{ cm}^3}$$ $$Q_t = V_g \cdot n = (8.462 \times 10^{-6}) \times 58.333 = 4.936 \times 10^{-4} \text{ m}^3/\text{s} = \mathbf{29.62 \text{ L/min}}$$ $$Q = Q_t \cdot \eta_v = (4.936 \times 10^{-4}) \times 0.92 = 4.541 \times 10^{-4} \text{ m}^3/\text{s} = \mathbf{27.25 \text{ L/min}}$$ Mechanical Efficiency: $$\eta_t = \eta_v \cdot \eta_m \implies \eta_m = \frac{\eta_t}{\eta_v} = \frac{0.89}{0.92} = 0.9674$$ $$\Delta P = 20 \times 10^6 - 0.2 \times 10^6 = 19.8 \times 10^6 \text{ Pa}$$ $$T = \frac{V_g \cdot \Delta P}{2\pi \cdot \eta_m} = \frac{(8.462 \times 10^{-6}) \times (19.8 \times 10^6)}{2\pi \times 0.9674} = \frac{167.55}{6.0784} = \mathbf{27.56 \text{ N}\cdot\text{m}}$$ $$N_{in} = T \cdot 2\pi n = 27.56 \times 2\pi \times 58.333 = 10,100 \text{ W} = \mathbf{10.10 \text{ kW}}$$

(b) Leakage Flow Rate & Leakage Resistance:
$$Q_L = Q_t - Q = 4.936 \times 10^{-4} - 4.541 \times 10^{-4} = 3.95 \times 10^{-5} \text{ m}^3/\text{s} = \mathbf{2.37 \text{ L/min}}$$ $$R_L = \frac{\Delta P}{Q_L} = \frac{19.8 \times 10^6}{3.95 \times 10^{-5}} = \mathbf{5.01 \times 10^{11} \text{ Pa}\cdot\text{s/m}^3} = \mathbf{501 \text{ GNs/m}^5}$$

(c) Real Flow & Driving Torque at $P = 28 \text{ MPa}$:
New pressure rise: $\Delta P_{28} = 28 \times 10^6 - 0.2 \times 10^6 = 27.8 \times 10^6 \text{ Pa}$.
$$Q_{L,28} = \frac{\Delta P_{28}}{R_L} = \frac{27.8 \times 10^6}{5.01 \times 10^{11}} = 5.55 \times 10^{-5} \text{ m}^3/\text{s}$$ $$Q_{28} = Q_t - Q_{L,28} = 4.936 \times 10^{-4} - 0.555 \times 10^{-4} = 4.381 \times 10^{-4} \text{ m}^3/\text{s} = \mathbf{26.29 \text{ L/min}}$$ $$T_{28} = \frac{V_g \cdot \Delta P_{28}}{2\pi \cdot \eta_m} = \frac{(8.462 \times 10^{-6}) \times (27.8 \times 10^6)}{2\pi \times 0.9674} = \mathbf{38.70 \text{ N}\cdot\text{m}}$$

Question 12: Swash Plate Axial Piston Pump

Given Data:
$z = 9$, $d = 8\text{ mm} = 0.008\text{ m}$, $D = 35\text{ mm} = 0.035\text{ m}$, $\gamma = 20^\circ$, $n = 3200\text{ rpm} = 53.333\text{ rps}$,
$\eta_m = 0.93$, $\eta_h = 0.99$, $P_i = 0$, $R_L = 260\text{ GNs/m}^5 = 2.6 \times 10^{11}\text{ Pa}\cdot\text{s/m}^3$.

Step-by-Step Solution:

(a) Geometric Volume & Pressure-Flow Relationship:
In a swash plate pump, stroke length is $S = D \tan \gamma$: $$S = 0.035 \times \tan 20^\circ = 0.035 \times 0.36397 = 0.01274\text{ m} = 12.74\text{ mm}$$ $$V_g = z \cdot \left(\frac{\pi}{4} d^2\right) \cdot S = 9 \times \frac{\pi}{4}(0.008)^2 \times 0.01274 = 5.762 \times 10^{-6}\text{ m}^3 = \mathbf{5.76\text{ cm}^3/\text{rev}}$$ $$Q_t = V_g \cdot n = (5.762 \times 10^{-6}) \times 53.333 = 3.073 \times 10^{-4}\text{ m}^3/\text{s} = \mathbf{18.44\text{ L/min}}$$ The characteristic equation relating real flow Q and exit pressure P is: $$Q(P) = Q_t - \frac{P}{R_L} = 3.073 \times 10^{-4} - \frac{P}{2.6 \times 10^{11}} \quad [\text{m}^3/\text{s}]$$ At $P = 0$: $Q = 18.44\text{ L/min}$. At $P = 40\text{ MPa}$: $Q_L = \frac{40 \times 10^6}{2.6 \times 10^{11}} = 1.538 \times 10^{-4}\text{ m}^3/\text{s} = 9.23\text{ L/min} \implies Q = 9.21\text{ L/min}$.

(b) Total Pump Efficiency at $P = 20\text{ MPa}$:
$$Q_{L,20} = \frac{20 \times 10^6}{2.6 \times 10^{11}} = 7.692 \times 10^{-5}\text{ m}^3/\text{s} = 4.615\text{ L/min}$$
$$\eta_v = 1 - \frac{Q_{L,20}}{Q_t} = 1 - \frac{4.615}{18.44} = 1 - 0.2503 = 0.7497 = 74.97\%$$ $$\eta_t = \eta_v \cdot \eta_m \cdot \eta_h = 0.7497 \times 0.93 \times 0.99 = \mathbf{0.6903 = 69.03\%}$$

(c) Maximum Delivery Pressure if Line is Completely Closed:
If the outlet pipe is deadheaded ($Q = 0$), all fluid displaced by the pump leaks internally: $$Q = 0 \implies Q_t = \frac{P_{max}}{R_L} \implies P_{max} = Q_t \cdot R_L$$ $$P_{max} = (3.073 \times 10^{-4}) \times (2.6 \times 10^{11}) = 7.99 \times 10^7\text{ Pa} \approx \mathbf{80\text{ MPa} = 800\text{ bar}}$$

Summary for Question 12:

Displacement: **$V_g = 5.76\text{ cm}^3$** | Total Efficiency at 20 MPa: **$\eta_t = 69.03\%$**

Deadhead Pressure (No relief valve): **$P_{max} \approx 80\text{ MPa (800 bar)}$** ⚠️ *(Catastrophic burst pressure!).*